

9.2 Exercises

Answers to selected odd-numbered problems begin on page ANS-23.

In Problems 1–8, $\mathbf{r}(t)$ is the position vector of a moving particle. Graph the curve and the velocity and acceleration vectors at the indicated time. Find the speed at that time.

1. $\mathbf{r}(t) = t^2\mathbf{i} + \frac{1}{4}t^4\mathbf{j}; t = 1$
2. $\mathbf{r}(t) = t^2\mathbf{i} + \frac{1}{t^2}\mathbf{j}; t = 1$
3. $\mathbf{r}(t) = -\cosh 2t\mathbf{i} + \sinh 2t\mathbf{j}; t = 0$
4. $\mathbf{r}(t) = 2 \cos t\mathbf{i} + (1 + \sin t)\mathbf{j}; t = \pi/3$
5. $\mathbf{r}(t) = 2\mathbf{i} + (t - 1)^2\mathbf{j} + t\mathbf{k}; t = 2$
6. $\mathbf{r}(t) = t\mathbf{i} + t\mathbf{j} + t^3\mathbf{k}; t = 2$
7. $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}; t = 1$
8. $\mathbf{r}(t) = t\mathbf{i} + t^3\mathbf{j} + t\mathbf{k}; t = 1$
9. Suppose $\mathbf{r}(t) = t^2\mathbf{i} + (t^3 - 2t)\mathbf{j} + (t^2 - 5t)\mathbf{k}$ is the position vector of a moving particle. At what points does the particle pass through the xy -plane? What are its velocity and acceleration at these points?
10. Suppose a particle moves in space so that $\mathbf{a}(t) = \mathbf{0}$ for all time t . Describe its path.
11. A shell is fired from ground level with an initial speed of 480 ft/s at an angle of elevation of 30° . Find:
 - (a) a vector function and parametric equations of the shell's trajectory,
 - (b) the maximum altitude attained,
 - (c) the range of the shell, and
 - (d) the speed at impact.
12. Rework Problem 11 if the shell is fired with the same initial speed and the same angle of elevation but from a cliff 1600 ft

high.

13. A used car is pushed off an 81-ft-high sheer seaside cliff with a speed of 4 ft/s. Find the speed at which the car hits the water.
14. A small projectile is launched from ground level with an initial speed of 98 m/s. Find the possible angles of elevation so that its range is 490 m.
15. A football quarterback throws a 100-yd “bomb” at an angle of 45° from the horizontal. What is the initial speed of the football at the point of release?
16. If a projectile is launched from level ground, then the initial conditions are $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{r}'(0) = \mathbf{v}_0 = (v_0 \cos \theta)\mathbf{i} + (v_0 \sin \theta)\mathbf{j}$ and the vector function (1) becomes

$$\mathbf{r}(t) = (v_0 \cos \theta)t\mathbf{i} + \left[-\frac{1}{2}gt^2 + (v_0 \sin \theta)t\right]\mathbf{j}. \quad (6)$$

This function is equivalent to the solution $x(t)$, $y(t)$ of the system of linear differential equations (10) in Problem 21 of Exercises 4.6 subject to the same initial conditions.

Underlying that earlier discussion, as well as in the derivation of (1), is the assumption that the projectile is not subject to air resistance. Use (6) to show that the impact speed of a projectile is the same as the initial speed $\|\mathbf{r}'(0)\| = \|\mathbf{v}_0\| = v_0$ for any angle of elevation θ , $0^\circ < \theta < 90^\circ$.

17. When air resistance is ignored, we saw in Problem 21 of Exercises 4.6 that a projectile launched at an angle θ , $0^\circ < \theta < 90^\circ$ from level ground ($\mathbf{r}(0) = \mathbf{0}$), then its horizontal range and maximum height are, respectively,

$$R = \frac{v_0^2}{g} \sin 2\theta \quad \text{and} \quad H = \frac{v_0^2}{2g} \sin^2 \theta.$$

From the first formula, it follows that when the projectile is launched at distinct complementary angles the horizontal

range is the same and that the maximum range is attained when the angle of elevation is $\theta = 45^\circ$. But if the projectile is launched from an initial height $s_0 > 0$ the foregoing formulas and statements are not valid.

- (a) With $v_0 = 480$ ft/s, $s_0 = 1600$ ft rework Problem 12, but this time use the complementary angle of elevation $\theta = 60^\circ$. Compare the maximum height, range, and impact speed of the projectile with the answers to parts (b), (c) and (d) of Problem 12.
- (b) Use a graphing utility or CAS to plot the trajectory of the projectile defined by the parametric equations $x(t)$ and $y(t)$ in part (a) of Problem 12. Repeat for the parametric equations in part (a) of this problem. Superimpose both of these curves on the same coordinate system.

18. As mentioned in Problem 17, if a projectile is launched from an initial height $s_0 > 0$ the maximum range of the projectile is not attained using $\theta = 45^\circ$ as the angle of elevation.

- (a) A projectile is launched from an initial height of $v_0 = 480$ ft/s, $s_0 = 1600$ ft, at an angle of elevation of $\theta = 45^\circ$. Use a calculator or CAS to find the time the projectile hits the ground and the corresponding range.
- (b) If the angle of elevation in part (a) is changed to $\theta = 39.76^\circ$, show that the range is greater than that in part (a).
- (c) Use a graphing utility or CAS to plot the trajectories of the projectiles in parts (a) and (b). Superimpose both of these curves on the same coordinate system.

19. If a projectile is launched from level ground, the initial conditions are $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{r}'(0) = \mathbf{v}_0 = (v_0 \cos \theta)\mathbf{i} + (v_0 \sin \theta)\mathbf{j}$, and if linear air resistance is taken into consideration, the vector function analogue of (1) is

$$\mathbf{r}(t) = \frac{mv_0 \cos \theta}{\beta} (1 - e^{-\beta t/m}) \mathbf{i} + \left[\left(\frac{mv_0 \sin \theta}{\beta} + \frac{m^2 g}{\beta^2} \right) (1 - e^{-\beta t/m}) - \frac{mg}{\beta} t \right] \mathbf{j}.$$

This function is equivalent to the solution $x(t)$, $y(t)$ of the system of linear differential equations (13) in Problem 22 of Exercises 4.6 subject to the same initial conditions. Here $\beta > 0$ is a constant of proportionality related to the air resistance or drag. If $m = \frac{1}{4}$ slug, $g = 32$ ft/s², $\beta = 0.02$, $v_0 = 300$ ft/s, and $\theta = 38^\circ$, use a calculator or CAS to find the impact speed of the projectile. See part (b) of Problem 22 in Exercises 4.6.

20. Suppose the angle of elevation of the projectile in Problem 19 is changed to $\theta = 52^\circ$. Do you think that the impact speed of the projectile is the same, greater than, or less than the value found in that problem? Prove your assertion.
21. A projectile is fired from a cannon directly at a target that is dropped from rest simultaneously as the cannon is fired. Show that the projectile will strike the target in midair. See **FIGURE 9.2.6**. [Hint: Assume that the origin is at the muzzle of the cannon and that the angle of elevation is θ . If $\mathbf{r}_p$ and $\mathbf{r}_t$ are position vectors of the projectile and target, respectively, is there a time at which $\mathbf{r}_p = \mathbf{r}_t$?

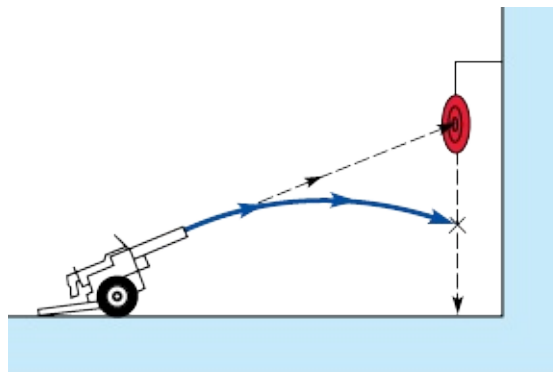


FIGURE 9.2.6 Cannon in Problem 21

22. In army field maneuvers sturdy equipment and supply packs

are simply dropped from planes that fly horizontally at a slow speed and a low altitude. A supply plane flies horizontally over a target at an altitude of 1024 ft at a constant speed of 180 mi/h. Use (1) to determine the horizontal distance a supply pack travels relative to the point from which it was dropped. At what line-of-sight angle α should the supply pack be released in order to hit the target indicated in **FIGURE 9.2.7**?

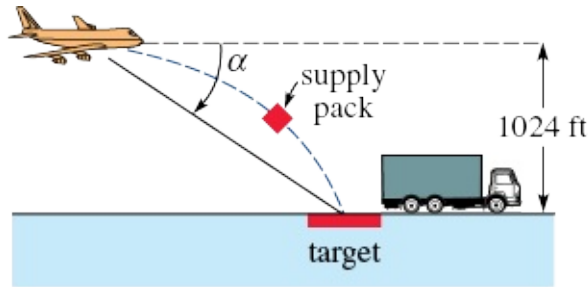


FIGURE 9.2.7 Supply plane in Problem 22

23. Suppose that $\mathbf{r}(t) = r_0 \cos \omega t \mathbf{i} + r_0 \sin \omega t \mathbf{j}$ is the position vector of an object that is moving in a circle of radius r_0 in the xy -plane. If $\|\mathbf{v}(t)\| = v$, show that the magnitude of the centripetal acceleration is $a = \|\mathbf{a}(t)\| = v^2/r_0$.

24. The motion of a particle in space is described by

$$\mathbf{r}(t) = b \cos t \mathbf{i} + b \sin t \mathbf{j} + ct \mathbf{k}, \quad t \geq 0.$$

(a) Compute $\|\mathbf{v}(t)\|$.

(b) Compute $s = \int_0^t \|\mathbf{v}(t)\| dt$ and verify that ds/dt is the same as the result of part (a).

(c) Verify that $d^2s/dt^2 \neq \|\mathbf{a}(t)\|$.

25. The **effective weight** w_e of mass m at the equator of the Earth is defined by $w_e = mg - ma$, where a is the magnitude of the centripetal acceleration given in Problem 23. Determine the effective weight of a 192-lb person if the radius of the Earth is 4000 mi, $g = 32 \text{ ft/s}^2$, and $v = 1530 \text{ ft/s}$.

26. Consider a bicyclist riding on a flat circular track of radius r_0 . If m is the combined mass of the rider and bicycle, fill in the blanks in **FIGURE 9.2.8**. [Hint: Use Problem 23 and force = mass $\times$ acceleration. Assume that the directions are upward and to the left.] The resultant vector $\mathbf{U}$ gives the direction the bicyclist must be tipped to avoid falling. Find the angle ϕ from the vertical at which the bicyclist must be tipped if her speed is 44 ft/s and the radius of the track is 60 ft.

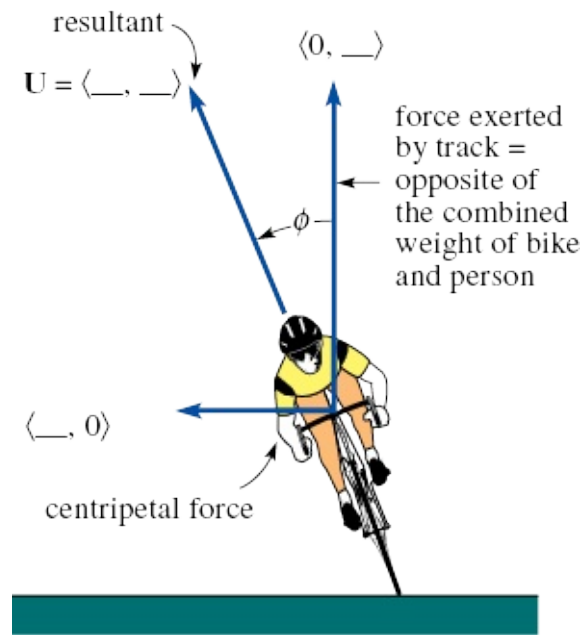


FIGURE 9.2.8 Bicyclist in Problem 26

27. The velocity of a particle moving in a fluid is described by means of a **velocity field** $\mathbf{v} = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$, where the components v_1 , v_2 , and v_3 are functions of x , y , z , and time t . If the velocity of the particle is $\mathbf{v}(t) = 6t^2x\mathbf{i} - 4ty^2\mathbf{j} + 2t(z + 1)\mathbf{k}$, find $\mathbf{r}(t)$. [Hint: Use separation of variables.]
28. Suppose m is the mass of a moving particle. Newton's second law of motion can be written in vector form as

$$\mathbf{F} = m\mathbf{a} = \frac{d}{dt}(m\mathbf{v}) = \frac{d\mathbf{p}}{dt},$$

where $\mathbf{p} = m\mathbf{v}$ is called **linear momentum**. The angular momentum of the particle with respect to the origin is defined to be $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, where $\mathbf{r}$ is its position vector. If the torque of the particle about the origin is $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times d\mathbf{p}/dt$, show that $\boldsymbol{\tau}$ is the time rate of change of angular momentum.

29. Suppose the Sun is located at the origin. The gravitational force $\mathbf{F}$ exerted on a planet of mass m by the Sun of mass M is

$$\mathbf{F} = -k \frac{Mm}{r^2} \mathbf{u}.$$

$\mathbf{F}$ is a **central force**—that is, a force directed along the position vector $\mathbf{r}$ of the planet. Here k is the gravitational constant, $r = \|\mathbf{r}\|$, $\mathbf{u} = \mathbf{r}/r$ is a unit vector in the direction of $\mathbf{r}$, and the minus sign indicates that $\mathbf{F}$ is an attractive force—that is, a force directed toward the Sun. See **FIGURE 9.2.9**.

- (a) Use Problem 28 to show that the torque acting on the planet due to this central force is $\mathbf{0}$.
- (b) Explain why the angular momentum $\mathbf{L}$ of a planet is constant.

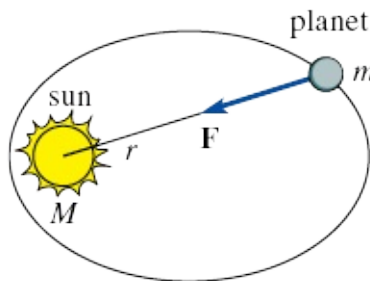


FIGURE 9.2.9 Force $\mathbf{F}$ in Problem 29

Discussion Problems

30. In this problem the student will use the properties in Sections

7.4 and 9.1 to prove **Kepler's first law of planetary motion**:

The orbit of a planet is an ellipse with the Sun at one focus. We assume that the Sun has mass M and is located at the origin, $\mathbf{r}$ is the position vector of a body of mass m moving under the gravitational attraction of the Sun, and $\mathbf{u} = \mathbf{r}/r$ is a unit vector in the direction of $\mathbf{r}$.

- (a) Use Problem 29 and Newton's second law of motion $\mathbf{F} = m\mathbf{a}$ to show that

$$\frac{d^2\mathbf{r}}{dt^2} = -kM \frac{\mathbf{u}}{r^2}.$$

- (b) Use part (a) to show that $\mathbf{r} \times \mathbf{r}'' = \mathbf{0}$.

- (c) Use part (b) to show that $\frac{d}{dt}(\mathbf{r} \times \mathbf{v}) = \mathbf{0}$.

- (d) It follows from part (c) that $\mathbf{r} \times \mathbf{v} = \mathbf{c}$, where $\mathbf{c}$ is a constant vector. Show that $\mathbf{c} = r^2(\mathbf{u} \times \mathbf{u}')$.

- (e) Show that $\frac{d}{dt}(\mathbf{u} \cdot \mathbf{u}) = 0$ and consequently $\mathbf{u} \cdot \mathbf{u}' = 0$.

- (f) Use parts (a), (e), and (d) to show that

$$\frac{d}{dt}(\mathbf{v} \times \mathbf{c}) = kM \frac{d\mathbf{u}}{dt}.$$

- (g) By integrating the result in part (f) with respect to t , we get $\mathbf{v} \times \mathbf{c} = kM\mathbf{u} + \mathbf{d}$, where $\mathbf{d}$ is another constant vector. Dot both sides of this last expression by the vector $\mathbf{r} = r\mathbf{u}$ and use Problem 61 in Exercises 7.4 to show that

$$r = \frac{c^2/kM}{1 + (d/kM) \cos \theta},$$

where $c = \|\mathbf{c}\|$, $d = \|\mathbf{d}\|$, and θ is the angle between $\mathbf{d}$ and $\mathbf{r}$.

- (h) Explain why the result in part (g) proves Kepler's first law.

(i) At perihelion, the point in the orbit where the body is closest to the Sun, the vectors $\mathbf{r}$ and $\mathbf{v}$ are perpendicular and have magnitudes r_0 and v_0 , respectively. Use this information and parts (d) and (g) to show that $c = r_0 v_0$ and $d = r_0 v_0^2 - kM$.

31. Suppose a projectile is launched from an initial height s_0 at an angle of elevation θ . If air resistance is ignored, show that the horizontal range is given by

$$R = \frac{v_0 \cos \theta}{g} (v_0 \sin \theta + \sqrt{v_0^2 \sin^2 \theta + 2s_0 g}).$$

Note that when $s_0 = 0$ this formula reduces to the range R given in Problem 17.

32. Consider the formula in Problem 31 as a function of the single variable θ . Show that the range of a projectile launched from an initial height s_0 is a maximum for the angle of inclination

$$\theta = \cos^{-1} \sqrt{\frac{2s_0 g + v_0^2}{2s_0 g + 2v_0^2}}.$$

Note that when $s_0 = 0$ this formula reduces to the value given in Problem 17, that is, $\theta = \pi/4$ or 45° . Use this formula to verify the angle of inclination used in part (b) of Problem 18.

* A projectile is shot or hurled rather than self-propelled. In the analysis of *long-range* ballistic motion, the curvature of the Earth must be taken into consideration.